Datasets and literature survey - Minimizer-Based Mutation Finder

Prepared 25 Oct 2026. Papers are 2024 or later.

Datasets and tools

What	Used for	Link	Status
Reference genome NC_045512.2 (SARS-CoV-2 Wuhan-Hu-1, about 29.9 kb)	Reference for both the project and the paper	https://www.ncbi.nlm.nih.gov/nuccore/NC_045512.2	Proposed. Confirm accession and length on NCBI.
Indian SARS-CoV-2 genomes (filter Geographic Region to India)	Paper only: real sample and answer key	NCBI Virus - SARS-CoV-2	No specific genome chosen. Accession numbers needed.
GISAID (largest Indian collection, INSACOG submissions)	Paper only, optional	https://gisaid.org	Needs registration and data terms.
Minimap2 (aligner)	Paper only: build the answer key, baseline comparison	https://github.com/lh3/minimap2	From memory; verify.

Literature survey (2024 and later)

Paper	Venue, year	Link	Why it matters here	Checked
1. Finimizers: Variable-Length Bounded-Frequency Minimizers for k-mer Sets Alanko, Biagi, Puglisi	**IEEE TCBB** 22(2), 2025	https://doi.org/10.1109/tcbbio.2025.3545285	Very frequent words slow index queries (long chains in our hash table); bounds the query time.	Preprint page opened; IEEE details from search result
2. Efficient seeding for error-prone sequences with SubseqHash2	Bioinformatics 41(8), 2025	https://academic.oup.com/bioinformatics/article/41/8/btaf418/8211825	Seeding that tolerates errors, compared against minimizers and syncmers.	Search result only
3. Fast genomic read alignment with minibwa Li, Homer	arXiv preprint, June 2026	https://arxiv.org/abs/2606.15357	Current fast short-read aligner; shows what the seed, chain, extend pipeline looks like today.	Page opened
4. Minimizing the Minimizers via Alphabet Reordering Verbeek, Ayad, Loukides, Pissis	arXiv, May 2024	https://arxiv.org/abs/2405.04052	The word ordering (our scrambling step) decides how many words are kept; the best order is NP-hard to find.	Page opened
5. Expected Density of Random Minimizers Golan, Shur	arXiv, Oct 2024	https://arxiv.org/abs/2410.16968	Predicts the fraction of words kept (slightly below 2/(w+1)); compare with our measured counts.	Page opened
6. Super Bloom: Fast and precise filter for streaming k-mer queries	bioRxiv preprint, 2026	https://www.biorxiv.org/content/10.64898/2026.03.17.712354v1.full	Bloom filter built for k-mer lookups; relates to our pre-check stage. Not peer reviewed.	Search result only
7. Recommendations for Uniform Variant Calling of SARS-CoV-2 Genome Sequence across Bioinformatic Workflows Connor et al.	Viruses, 2024	https://doi.org/10.3390/v16030430	Calling thresholds and workflow choices for SARS-CoV-2; context for our pile-up rule.	Search result only
8. Investigating Sensitivity, Specificity and Accuracy of Variant Calling Pipelines for Analyzing SARS-CoV-2 Data	bioRxiv preprint, 2024	https://www.biorxiv.org/content/10.1101/2024.01.24.576385v1.full	Compares mapper and caller combinations. Not peer reviewed.	Search result only
9. A Benchmark of methods for SARS-CoV-2 whole genome sequencing and development of a more sensitive method Bayega et al.	Frontiers in Genetics 16, 2025	https://www.frontiersin.org/journals/genetics/articles/10.3389/fgene.2025.1516791/full	Low viral load gives low coverage; guides our coverage and depth choices.	Page opened
10. SARS-CoV-2 sequencing artifacts associated with targeted PCR enrichment and read mapping	PLOS One, 2025	https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0334009	Ambiguous (N) bases caused by read-mapping artifacts; supports our N handling.	Search result only

Only paper 1 is from IEEE (shaded); no other IEEE paper from 2024 onward turned up in my searches, so run the IEEE Xplore search yourself too. "Page opened" means the paper's page was read to confirm title, authors and date. "Search result only" means the details come from a search listing; open each paper before citing it.